

Episcleral brachytherapy in Portugal for the treatment of uveal melanoma

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Abstract

Purpose: To evaluate the outcomes of our institution, the National Ocular Oncology Reference Center (Centro Hospitalar e Universitário de Coimbra) on Episcleral Brachytherapy (EBT) for the treatment of uveal melanoma (UM), since its beginning in November 2013, regarding local control, survival rates, metastatic rates, and side effects and to compare them with the international literature.

Materials and Methods: Prospective study of patients referred to the Ocular Oncology Reference Center and treated with EBT between November 2013 and August 2020. Data were collected regarding local control, survival, distant metastasis, and treatment side effects. Data analysis evaluated treatment outcomes and radiation side effects.

Results: A total of 150 patients underwent EBT but only 143 were considered for analysis. In 95.5% of cases local control was achieved and only 4.5% underwent secondary enucleation due to treatment failure and disease progression. Tumor initial maximum diameter was a predictor of time to melanoma-related death (Hazard ratio [HR] = 1.24 [95% confidence interval [CI] - 1.03–1.50] $P = 0.018$) and time to metastasis (HR = 1.43 [95% CI - 1.2–1.71] $P \leq 0.001$). Among the 143 patients, the most common radiation side effect was cataract (58.33%) followed by the development of any form of radiation retinopathy (42.36%). One patient (0.7%) underwent secondary enucleation due to uncontrolled neovascular glaucoma.

Conclusions: Our results demonstrate excellent clinical outcomes of local control with EBT, with satisfactory overall survival and metastasis-free survival rates, comparable to international literature. Considering the high rates of local control and globe preservation, this research validates the use of EBT as a mainstay treatment in UM.

Keywords: Episcleral brachytherapy, outcomes, uveal melanoma

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INTRODUCTION

Ocular melanoma is the second most common type of melanoma and is the most common primary intraocular malignancy in adults.^[1] It arises from melanocytes in the

conjunctiva and uveal tract, most frequently from choroidal melanocytes (85%–90%), but also from the iris (3%–5%) and ciliary body (5%–8%).^[2,3]

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In Europe, standardized incidence rates vary from <2 cases/1 000 000 in Southern countries, such as Spain and Italy, to more than 8/1 000 000 in Northern European Countries.^[4] The median age of diagnosis is approximately 62 years, peaking between 70 and 79 years.^[3] Numerous studies have evaluated the factors associated with increased risk of uveal melanoma (UM). Currently, accepted risk factors include age, race (fair skinned individuals with Northern European ancestry), choroidal nevi or melanocytoma, ocular melanocytosis and *BAP-1* predisposition syndrome (germline *BAP-1* mutations).^[3,5] *BAP-1* is a tumor-suppressor gene located on chromosome 3, that is mutated in 47% of UM.^[3]

UM diagnosis is mostly based on clinical examination and confirmed by diagnostic techniques such as ultrasonography (US) and optical coherence tomography (OCT).^[5,6]

On fundus examination, melanomas may be pigmented (55%), nonpigmented (15%) or mixed (30%) and may appear in one of three configurations: dome-shaped (75%), mushroom-shaped (20%) (breaking through Bruch's membrane and herniating into the subretinal space) or diffuse (5%) (flat lesions).^[5]

US is important for diagnostic confirmation, classically showing low to medium reflectivity on A-scan and acoustically hollowness on B-scan.^[6] OCT is best used for small UM (<3 mm in thickness) and shows a dome-shaped choroidal mass, compressing the overlying choriocapillaris. This imaging modality is especially useful to demonstrate the presence of subretinal fluid and cystoid macular edema.

Early diagnosis and treatment of UM are essential since survival rates correlate with primary tumor size and up to 50% of patients develop metastatic disease (through hematogenous dissemination).^[3,5] The *Collaborative Ocular Melanoma Study* (COMS) reported a 5- and 10-year cumulative metastasis rates of 25% and 34%.^[5,9] The liver is the most common site of metastasis and is involved in up to 90% of cases.^[5]

Predictive factors for metastatic disease include tumor diameter, involvement of the ciliary body, extrascleral extension, epithelioid melanoma cytology, high mitotic rate, and chromosome mutations.^[5] The American Joint Committee on Cancer (AJCC) T1–T4 staging system is based on some of these characteristics and survival rates decrease rapidly with increasing stage^[3] (APPENDIX).

In the early 1900s, patients often presented with large UM for which the primary treatment was enucleation.

However, even after elimination of the primary tumor, there was high mortality due to metastasis. Zimmerman *et al.* proposed that a spike in intraocular pressure, at the time of the optic nerve cut, could cause dissemination of tumor cells through the vortex veins into the systemic circulation.^[6] The Zimmerman theory led, in 1970s, to pursuit globe-preserving treatments that could prevent metastatic dissemination and also maintain visual function, prompting the development of episcleral brachytherapy (EBT).^[7] This method consists in suturing onto the sclera curvilinear plaques containing radioactive isotopes such as iodine-125 (¹²⁵I), ruthenium-106 (¹⁰⁶Ru), iridium-192 (¹⁹²Ir), and palladium-103 (¹⁰³Pd)^[9] to deliver trans-scleral radiation to UM.

Two multicenter randomized clinical trials from the COMS Group established EBT as the mainstay of treatment. The trial for medium-sized melanomas showed no significant differences in overall survival rates at 5 and 12 years between groups randomized to enucleation or EBT.^[8,9] A second trial for large melanomas showed similar 5- and 10-year cumulative tumor-related mortality rates between groups randomized to enucleation or preenucleation external-beam radiation.^[10,11]

However, this procedure may cause iatrogenic radiation damage to ocular tissues, commonly causing radiation retinopathy (RR). Pathologic studies of RR demonstrate endothelial cell loss and capillary closure with vitreoretinal neovascularization in the presence of severe retinal ischemia.^[12] Efforts to reduce the incidence of these radiation-induced complications led to the widespread use of low-emitting isotopes such as iodine-125^[13] and more recently, prophylactic intravitreal bevacizumab injections for the prevention of radiation side effects.^[14]

The efficacy of EBT has been worldwide established in major institutions. The purpose of this scientific research is to compare those results with the outcomes of our Institution, since its beginning in November 2013, regarding local control, survival rates, metastatic rates, and side effects.

MATERIALS AND METHODS

Study design

Prospective case series including 150 patients treated with ¹²⁵I EBT for UM at the Ocular Oncology Portuguese Reference Center, at Centro Hospitalar e Universitário de Coimbra between November 2013 and August 2020. Based on the COMS classification system and the guidelines from the American Brachytherapy Society,^[15] EBT treatment was offered to:

- Patients with medium-sized melanomas
- Patients with small melanomas or suspicious choroidal pigmented lesions with documented growth and/or presence of more than three risk factors for transformation into UM^[16]
- Some large melanomas with potential for visual conservation, provided that EBT available plaques could adequately circumscribe the base of the tumor with adequate safety margins;

EBT Exclusion Criteria:

- Circumpapillary or peripapillary melanomas that could not be correctly irradiated with EBT (patients are offered proton beam irradiation)
- Large-sized melanomas in blind painful eyes or with no potential for visual conservation (patients are offered primary globe enucleation)
- UMs with extra-ocular extension >2 mm and no possibility of adequate irradiation
- Evidence of metastatic UM or any other primary cancer

All patients were fully informed about the treatment options and potential adverse effects and gave written consent.

Clinical evaluation and data collection

In all patients, a complete ophthalmological evaluation was performed including, best corrected visual acuity (BCVA), comprehensive slit-lamp evaluation, measurement of intraocular pressure (Goldmann tonometry), and dilated fundus examination. All patients performed additional ancillary exams including fundus color photography (Nikon Digital SLR Camera D7000 [Nikon Corporation, Japan] mounted on a TRC-NW7SF Mark II Retinal Camera [TopCon Corporation, Japan]) and measurements of tumor dimensions using B-mode US with vector A (Ultra Scan Imaging System™ and UBM Plus-P40™, Paradigm, Medical Industries, Inc., USA). Spectral domain OCT (standard deviation [SD]-OCT) (Heidelberg Engineering, Germany) was used to scan UM and document the presence of intraretinal or subretinal fluid in all patients.

Systemic evaluation was completed with laboratory testing (complete blood count, liver and renal function markers) and imaging (abdominal ultrasound, thoracic and abdominal computed tomography, or hepatic magnetic resonance imaging).

Collected data included general demographic, UM location, largest basal diameter and thickness, staging (COMS and AJCC), BCVA at baseline and during all follow-up visits,

plaque type and size, follow-up time, radiation doses to ocular structures, development of EBT adverse effects, UM metastasis, and local treatment failure. Treatment failure was defined by any degree of enlargement of the residual tumor in base or thickness detected by ophthalmoscopy or US or the presence of extrascleral extension >2 mm. In all of these cases, secondary enucleation was offered to the patient.

Treatment protocol

Using a treatment planning software, Plaque Simulator® (version 5.3.9, *Eye Physics LLL, EUA*) a tridimensional reconstruction (based on ophthalmological examination and imaging exams) of the tumor and adjacent ocular structures was accomplished. A prescription dose of 85 Gy to the tumor apex was planned taking into consideration the duration of treatment, the plaque size, number and distribution of ¹²⁵I seeds, the need for UM irradiation with a 2 mm margin and the amount of radiation affecting adjacent ocular structures. Radiation doses to ocular tissues such as sclera, optic nerve, macula, and lens were registered.

Different sized COMS-type plaques (*IBT BEBIG, Inc*) and ROPES plaques (*Radiation Oncology Physics and Engineering Services Ltd, Australia*) with ¹²⁵I seeds (*IBT BEBIG I25, S16, classes A04 to A14*) were used.

All procedures were performed with general anesthesia. The plaque was sutured with Vicryl 5/0 (polyglactin 910, Ethicon Inc) to the sclera underlying the tumor. This procedure was assisted by pupillary transillumination (marking the base of the melanoma to find the correct plaque position) and further confirmed with intraoperative US. If needed, temporary extraocular muscle disinsertion was performed.

After surgery, the patient remained in an isolated room with radioactive protection during the preestablished treatment period. Subsequently, a second surgical procedure for plaque extraction and, when necessary, extraocular muscle reinsertion was completed under general anesthesia.

Patients were seen according to the following protocol: every 2 weeks in the 1st month, every 3 months during the 1st year and every 6 months during the following 5 years. This protocol was personalized if needed in case of ocular or systemic complications. During this follow-up, complete blood work-up and abdominal imaging were requested every 6 months.

An ocular oncology trained ophthalmologist examined every patient each visit and recorded tumor dimensions (obtained by US), detecting cases of treatment failure and evaluating

acute and late signs of radiation toxicity. RR included both radiation maculopathy and radiation neuropathy. Radiation maculopathy was defined as retinal capillary bed changes (nonperfusion, microaneurysms, and retinal hemorrhages), retinal exudation, retinal edema, nerve fiber layer infarctions or vascular sheathing in the macular area. Radiation neuropathy was considered to be present if optic disc swelling, hemorrhages, and peri-papillary exudation were observed.

Statistical analysis

Population demographics, clinical and imaging characteristics were summarized using traditional descriptive statistic methods.

The primary outcome of this study was treatment failure, defined by any degree of enlargement of the irradiated tumor in base or thickness, or extrascleral extension). Mortality rates (overall survival) and metastatic disease (metastasis-free survival) were defined as the secondary outcomes.

For the primary and secondary outcomes, incidence proportions were calculated. The change from baseline to last follow-up regarding tumor dimensions and BCVA were defined as exploratory outcomes.

In order to test which demographic and tumor characteristics or features of brachytherapy could predict the risk of treatment failure, overall survival and metastasis-free survival, Cox proportional-Hazard Models were built. Each predictor was tested on separate univariate models and hazard ratios (HRs) with 95% confidence interval (CI) are reported.

Changes from baseline to last follow-up of continuous exploratory outcomes were compared with paired *t*-tests.

All statistics were performed on STATA (version 16.1, StataCorp LCC, College Station, TX, USA) and $P < 0.05$ values were considered statistically significant.

RESULTS

A total of 150 patients with UM underwent ^{125}I EBT between November 2013 and August 2020; 145 completed a minimum follow-up of 2 months postirradiation and 2 patients were lost to follow-up. Mean follow-up duration was 29.88 months (SD = 19.79; range 2–76.4).

The mean age at diagnosis was 61.31 years (SD = 13.38; range 26–87 years) with a slight predominance of females (55.94%) and right eyes (51.75%). Baseline

demographic and clinical characteristics are summarized in Table 1.

Mean treatment duration was 5.97 days (SD = 1.63; range 3–10), with a prescribed dose of 85 Gy to the tumor apex. Dosimetric data to adjacent ocular structures are provided in Table 2. One hundred and thirteen COMS plaques and 31 ROPES plaques were used, with a median size of 15 mm and ranging between 12 and 20 mm.

Baseline mean diameter was 11.34 mm (SD = 2.95; range 3.1–18.93) and baseline mean thickness 6.12 mm (SD = 2.21; range 2–13). Tumor classification

Table 1: Baseline demographics, clinical evaluation and tumor characteristics

Demographics	
Age (years)	61.31±13.38
Sex (female) (%)	80/143 (55.94)
Eye (right) (%)	74/143 (51.75)
Follow-up (months) (mean)	29.88±19.79 (2-76.74)
Clinical variables	
Visual acuity (logMAR)	0.88±0.91
Tumor type (%)	
Choroidal	129/143 (90.21)
Ciliary body	12/143 (8.39)
Iris and ciliary body	2/143 (1.40)
Tumor location (%)	
Posterior to the equator	53/143 (37.06)
Macular	45/143 (31.47)
Anterior to the equator	20/143 (13.99)
Peripapillary	13/143 (9.09)
Ciliary body (± Iris)	12/143 (8.39)
Tumor diameter (mm)	11.34±2.95
Tumor thickness (mm)	6.12±2.21
AJCC stage (%)	
I	22/143 (15.38)
IIA	66/143 (45.15)
IIB	51/143 (35.66)
IIIA	4/143 (2.80)
COMS classification (%)	
I	22/143 (15.38)
II	117/143 (81.82)
III	4/143 (2.80)

LogMAR: Logarithm minimum angle of resolution, COMS: Collaborative ocular melanoma study, AJCC: American Joint Committee on Cancer

Table 2: Details of episcleral brachytherapy treatments

Treatment duration (days)	5.97±1.63
Plaque type (%)	
COMS	(112/143) 78.32
ROPES	(31/143) 21.68
Plaque size (median)	15 (12-20)
Radiation doses (Gy)	
Apex	94.63±15.85 (58.95-178.75)
Sclera	364.00±207.39 (61.42-1202.18)
Optic nerve	55.10±41.66 (6.85-346.13)
Lens	26.64±17.70 (4.70-115.0)
Macula	94.36±90.41 (6.59-533)

COMS: Collaborative ocular melanoma study, ROPES: Radiation Oncology Physics and Engineering Services

according to COMS and the AJCC staging systems is presented in Table 1.

Treatment failure and local control

In 95.5% of cases local control was achieved and only 4.5% underwent secondary enucleation due to treatment failure and disease progression, with a mean time of 23.08 ± 17.8 months. The 2-year and 5-year local control rates were 96.67% and 87.89%, respectively. Regarding tumor dimensions, there was a significant reduction in basal diameter ($t = -11.83$; $P < 0.001$) and thickness ($t = -12.26$; $P < 0.001$).

Our analysis also verified a significant reduction in BCVA ($t = 9.51$; $P < 0.001$). Figure 1 depicts a Kaplan–Meier estimate of patients with tumor local control after EBT.

Survival and systemic disease

At the end of follow up, 17 patients had developed confirmed or suspected melanoma metastasis. Survival estimates are shown below (Figure 2 overall survival and Figure 3 metastasis free survival). The 2-year and 5-year overall survival was 99.16% and 80.69%, respectively. Regarding metastasis free survival the 2-year and 5-year rates were 95.21% and 80.56%, respectively.

Treatment side-effects

Among the 143 patients considered for analysis, the most common radiation side-effect was cataract (58.33%), followed by the development of any form of RR (42.36%). *Rubeosis iridis* and neovascular glaucoma were also documented in 13.19% and 10.42% of the cases, respectively. For the treatment of RR, anti-VEGF intravitreal injections were administered, with a mean of 5 injections per patient (SD = 2.83; range 1–18). One patient underwent

secondary enucleation due to uncontrolled neovascular glaucoma.

Free RR rates at 2-year and 5-years were 83.78% and 35.32%, respectively.

Exploratory analysis of predictors of the outcomes

We used univariate Cox regression models to test for variables that may predict the primary (treatment failure) and secondary outcomes (overall survival and metastatic disease).

The only predictor with criteria of statistical significance for the primary endpoint (treatment failure) was ciliary body location (HR = 23.55 [95% CI - 5.57–99.42] $P \leq 0.001$) [Table 3].

When considering secondary outcomes, we aimed to explore whether any demographic, clinical or treatment variable (s) could predict survival outcomes. The only predictor that met the criteria of statistical significance on univariate analysis for time to death (HR = 1.24 [95% CI - 1.03-1.50] $P = 0.018$) and time to metastasis (HR = 1.43 [95% CI - 1.2–1.71] $P \leq 0.001$) was baseline tumor largest basal diameter.

Neither AJCC or COMS staging, BCVA, sex or tumor thickness showed significance as predictors [Tables 4 and 5].

DISCUSSION

The main purpose of EBT is the adequate tumor local control, preserving the globe, with a minimum of associated side effects. We hereby present and discuss our results regarding primary and secondary endpoints, evaluating EBT as globe conservative approach in treatment of UM.

Table 3: Univariate analysis of the treatment failure

	Univariate analysis	
	HR (95% CI)	P
Demographic variables		
Age	1.05 (0.99-1.11)	0.095
Male sex	0.73 (0.17-3.08)	0.675
Baseline visual acuity	1.19 (0.60-2.35)	0.612
Clinical variables		
Tumor type		
Choroidal	1	
Ciliary body	23.55 (5.57-99.42)	≤ 0.001
Iris+ciliary body	7.2 (0)	1
Location		
Posterior to equator	1	
Macular	0.51 (0.05-5.10)	0.572
Anterior to equator	2.22 (0.36-13.49)	0.383
Peripapillary	3.04 (0)	1
Ciliary body	4.73 (0.77-29.02)	0.043
Tumor baseline diameter	1.0 (0.78-1.30)	0.946
Tumor baseline thickness	1.12 (0.84-1.49)	0.408

CI: Confidence interval, HR: Hazard ratio

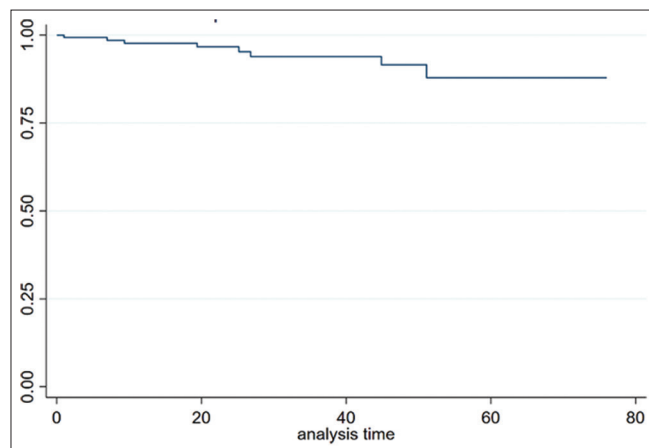


Figure 1: Kaplan–Meier estimates of patients with local control after episcleral brachytherapy

Local control and treatment failure

The primary outcome of this work was treatment failure and local control after treatment with ^{125}I . The 2-year and 5-year local control rates were 96.7% and 87.9%, respectively. A 2-year and 5-year recurrence rates of 3.3% and 12.1% were documented in our cohort.

Our results demonstrate excellent outcomes for local control rates comparable to other Institutions around the world and the COMS multicentric trial results.^[8-11]

Table 4: Univariate analysis of the time to death (secondary outcome)

	Univariate analysis	
	HR (95% CI)	P
Demographic variables		
Age	1.03 (0.99-1.07)	0.133
Male sex	0.55 (0.19-1.58)	0.268
Baseline visual acuity	0.99 (0.59-1.66)	0.976
Clinical variables		
Tumor type		
Choroidal	1	
Ciliary body	1.54 (0.2-11.95)	0.674
Iris plus ciliary body	4.76 (0.6-37.16)	0.136
Conjunctiva	2.15 (0)	1
Location		
Posterior to equator	1	
Macular	6.67 (1.3-32.16)	0.018
Anterior to equator	3.48 (0.48-25.1)	0.215
Peripapillary	4.01 (0.56-28.6)	0.166
Ciliary body	7.63 (1.06-54.47)	0.043
Tumor baseline diameter	1.24 (1.03-1.50)	0.018
Tumor baseline thickness	0.87 (0.69-1.10)	0.276
COMS classification		
I	1	
II	1.39 (0.18-10.9)	0.752
III	6.28 (0.38-104.1)	0.199
ACJJ stage		
I	1	
IIA	1.57 (0.19-12.8)	0.674
IIB	1.20 (0.13-10.3)	0.867
IIIA	6.24 (0.37-103.5)	0.201

CI: Confidence interval, HR: Hazard ratio, COMS: Collaborative ocular melanoma study

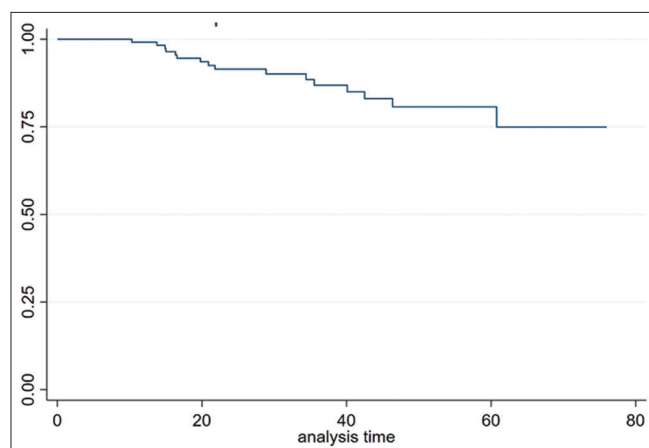


Figure 2: Kaplan–Meier estimates of overall survival

Only 4.5% underwent secondary enucleation due to treatment failure and disease progression. In 2002, a report from COMS evaluated the frequency of local relapse after treatment with ^{125}I to be 10.3%^[17] and a recent meta-analysis showed that the risk of local recurrence of UM treated with ^{125}I varied between 4.0% and 9.6%.^[17] Furthermore, a Spanish study with patients sharing similar characteristics with our population showed a 2-year local recurrence of 5.7% and a 5-year of 11.6%.^[19]

On our study, the only predictor of treatment failure was ciliary body tumor location ($P \leq 0.001$), usually associated with worse prognosis.^[5] According to a study by Kowal *et al.* one of the factors associated with local relapse was the involvement of the ciliary body.^[20] Kaliki and Shields confirmed that melanomas developing in the ciliary body or with secondary involvement of the ciliary body are more aggressive in their course than those which extend to the iris or choroid.^[21] On explanation is the late diagnosis and therefore larger dimensions increasing the risk of a local recurrence.^[20]

Correa *et al.* found a statistically significant association between treatment failure and a higher COMS stage.^[19] The 5-year COMS report for treatment failure and enucleation found that older age, greater tumor thickness and proximity to the foveal avascular zone were risk factors for treatment failure.^[17]

According to the Ophthalmic Oncology Task Force, there is a significant difference in survival after treatment of UM between groups of patients with local relapse and without relapse. 5-year and 10-year survival is 71% and 62% (local relapse) opposed to 87% and 82% (without relapse).^[20]

Survival and systemic disease

UM disseminates hematogenously with a high propensity to

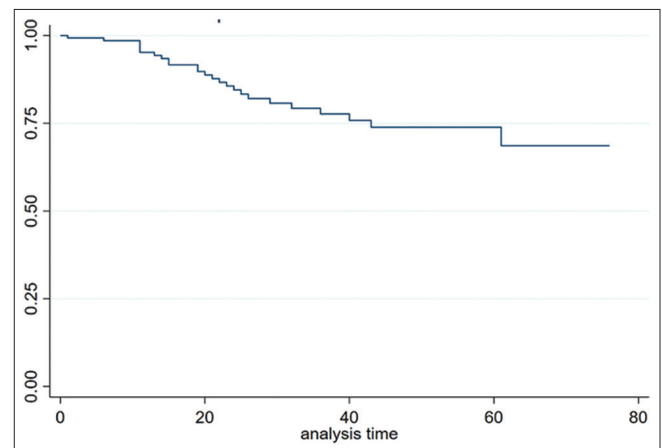


Figure 3: Kaplan–Meier estimates of overall metastasis-free survival after episcleral brachytherapy

Table 5: Univariate analysis of time to metastasis (secondary outcome)

	Univariate analysis	
	HR (95% CI)	P
Demographic variables		
Age	0.99 (0.95-1.02)	0.576
Male sex	0.73 (0.29-1.87)	0.523
Baseline visual acuity	1.06 (0.67-1.69)	0.773
Clinical variables		
Tumor type		
Choroidal	1	
Ciliary body	3.31 (0)	1.00
Iris plus ciliary body	3.34 (0)	1.00
Location		
Posterior to equator	1	
Macular	9.96 (1.24-79.71)	0.030
Anterior to equator	8.29 (0.86-79.98)	0.067
Peripapillary	20.86 (2.43-178.77)	0.006
Ciliary body	6.67 (0.41-107.23)	0.180
Tumor baseline diameter	1.43 (1.2-1.71)	≤0.001
Tumor baseline thickness	1.02 (0.84-1.25)	0.791
COMS classification		
I	1	
II	2.22 (0.29-16.87)	0.439
III	9.83 (0.61-158.21)	0.107
ACJJ stage		
I	1	
IIA	2.51 (0.31-19.81)	0.674
IIB	1.90 (0.23-15.63)	0.867
IIIA	9.84 (0.61-158.45)	0.201

CI: Confidence interval, HR: Hazard ratio, COMS: Collaborative Ocular Melanoma Study, AJCC: American Joint Committee on Cancer

create metastasis in the liver (93%), followed by lung (24%) and bone (16%).^[2,3] At the time of diagnosis, <4% of patients with UM have detectable metastatic disease.^[2] In a study by Shields *et al.*, metastasis were found in 15% and 25% of patients after 5 and 10 years of follow-up,^[22] respectively. After metastasis detection, 80% of patients die within 1 year and 92% within 2 years.^[2] Long term survivals are rare and mean survival after overt metastatic disease is only a few months.^[2,23] In our study 17 patients (12.5%) developed systemic disease, and from those only 4% are still alive.

Considering time to metastasis, we found that largest basal diameter was a strong predictor of time to metastasis. In our work the 2-year and 5-year metastasis free survival was 95.21% and 80.56%, respectively. Our results are comparable to the studies previously described.

In this study, the 2-year and 5-year overall survival was 99.16% and 80.69%, respectively. Regarding overall survival Correa *et al.* described a 2-year and 5-year rate of 94.4% and 84.1%, respectively.^[19] The COMS study reported a 5-year survival of 82% for patients undergoing treatment with EBT. The overall survival results of our cohort are similar.

When studying possible predictors of mortality, we found that largest basal tumor diameter was a strong predictor of

time-to-death. Increased tumor thickness and large basal diameter are well-defined predictors of poor prognosis, distant metastasis and melanoma-related death.^[2] Our results are comparable to several published studies in which greater tumor basal diameters positively correlated with melanoma-related death.

Treatment side-effects

Radiation treatment is associated with both early and late onset side-effects. Cataract is an early complication and its formation is related to multiple mechanisms such as deformation of heat labile enzymes, damage to cellular DNA and physical destruction of lens cells through thermoelastic expansion.^[24] Cataract risk varies according to tumor size and location. In the COMS trial the risk of cataract was 85% for anterior tumours and 17% for posterior tumours.^[25] Gunduz K *et al.* reported a 5-year probability of developing cataract of 32%.^[26] In our study a higher percentage was observed and the development of cataract was recorded as the most common side-effect after EBT (affected 58.33% of patients). One explanation for this fact may be related to the high prevalence of tumors occupying an anterior location on the eye (22% of tumors located anteriorly to the equator and involving ciliary body).

RR is a vision-threatening complication that tends to manifest later during follow-up. RR manifestations increase in prevalence during follow-up and prevalence rates for nonproliferative RR and radiation maculopathy have been reported in retrospective studies to range from 42% to 94%.^[12] However, in these studies there were no standardized definitions of RR, sources of radiation and tumor sizes. Another limitation is the difficulty of separating tumor effects from radiation effects, since it is not possible to have a control group for ethical reasons. Gündüz *et al.* reported a 40% risk of developing RR at 5 years and the most important predictors for the development of proliferative RR were diabetes, use of iridium-192 (a higher energy isotope than iodine-125) and a tumor base >10 mm.^[26]

We report free RR rates at 2-year and 5-years of 83.78% and 35.32%, respectively. At 5 years we have rates of RR higher than the aforementioned study. On possible explanation is our high percentage of macular and retro equatorial tumors (~67%). There is no approved treatment regimen for RR and large-scale randomized trials are lacking, however intravitreal bevacizumab has been showed to stabilize visual acuity loss and progressively reduce RR lesions and macular edema.^[27] Our patients were offered treatment with intravitreal bevacizumab injections (3-injections for loading dose followed by a

PRN regimen) with a mean of 5 injections per patient. Nonetheless, *rubeosis iridis* and neovascular glaucoma were documented in 13.19% and 10.42% of the cases, respectively, and secondary enucleation was offered to one patient due to refractory neovascular glaucoma. New studies are evaluating the outcomes of prophylactic bevacizumab to reduce RR rates.^[14]

This study has some limitations, particularly being developed at single institution with moderate follow-up time and including a relatively small number of patients when compared to tertiary centers in larger US and European countries. Moreover, some events have low absolute numbers, limiting inferential analysis of the data.

CONCLUSIONS

UM is a complex rare condition that requires a multidisciplinary approach and inter institutional collaboration. Our results demonstrate excellent clinical outcomes of local control with EBT, with satisfactory overall survival and metastasis-free survival rates. These results are comparable to the published international literature, showing that UM patients treated at the Portuguese Ocular Oncology Reference Centre are offered the standard of care. Considering the high rates of local control and globe preservation, this study validates the current use of EBT as a mainstay treatment for UM.

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Conflicts of interest

There are no conflicts of interest.

REFERENCES

- Patel DR, Patel BC. Cancer, ocular melanoma. In: StatPearls. Treasure Island (FL): StatPearls Publishing; 2020. [Last updated on 2020 May 24]. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK551647/>.
- Jovanovic P, Mihajlovic M, Djordjevic-Jocic J, Vlajkovic S, Cekic S, Stefanovic V. Ocular melanoma: An overview of the current status. *Int J Clin Exp Pathol* 2013;6:1230-44.
- Krantz BA, Dave N, Komatsubara KM, Marr BP, Carvajal RD. Uveal melanoma: Epidemiology, etiology, and treatment of primary disease. *Clin Ophthalmol* 2017;11:279-89.
- Fonseca C, Casalta-Lopes J, Teixeira T, Cavaco A, Simões P, Cachulo M, et al. Braquiterapia episcleral no tratamento do melanoma da úvea – A nossa experiência. *Oftalmologia* 2016; 40: 27-33.
- Singh M, Durairaj P, Yeung J. Uveal melanoma: A review of the literature. *Oncol Ther* 2018;6:87-104.
- Tarlan B, Kiratlı H. Uveal Melanoma: Current Trends in Diagnosis and Management. *Turk J Ophthalmol*. 2016 Jun;46(3):123-137. doi: 10.4274/tjo.37431. Epub 2016 Jun 6. PMID: 27800275; PMCID: PMC5076295.
- Brewington BY, Shao YF, Davidorf FH, Cebulla CM. Brachytherapy for patients with uveal melanoma: historical perspectives and future treatment directions. *Clin Ophthalmol*. 2018;12:925-934. Published 2018 May 17. doi:10.2147/OPTH.S129645
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: Initial Mortality Findings. COMS Report no. 18. *Arch Ophthalmol* 1998 Jun; 125(6): 779-796
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, V: Twelve-year mortality rates and prognostic factors. COMS Report no. 28. *Arch Ophthalmol* 2006; 124: 1684-1693
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of pre-enucleation radiation of large choroidal melanoma II: initial mortality findings. COMS Report no. 10. *Am J Ophthalmol* 2001 Jun; 119: 969-982
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of pre-enucleation radiation of large choroidal melanomas: IV: Ten-year mortality findings and prognostic factors. COMS Report no. 24. *Am J Ophthalmol* 2004; 138: 936-951
- Boldt HC, Melia BM, Liu JC, Reynolds SM; Collaborative Ocular Melanoma Study Group. I-125 brachytherapy for choroidal melanoma photographic and angiographic abnormalities: the Collaborative Ocular Melanoma Study: COMS Report No. 30. *Ophthalmology*. 2009 Jan;116(1):106-115.e1. doi: 10.1016/j.ophtha.2008.10.013. PMID: 19118701; PMCID: PMC3202984.
- Echegaray JJ, Bechrakis NE, Singh N, Bellerive C, Singh AD. Iodine-125 Brachytherapy for Uveal Melanoma: A Systematic Review of Radiation Dose. *Ocul Oncol Pathol*. 2017;3(3):193-198. doi:10.1159/000455872
- Chang M, Dalvin LA, Mazloumi M, et al. Prophylactic Intravitreal Bevacizumab After Plaque Radiotherapy for Uveal Melanoma: Analysis of Visual Acuity, Tumor Response, and Radiation Complications in 1131 Eyes Based on Patient Age. *Asia Pac J Ophthalmol (Phila)*. 2020;9(1):29-38. doi:10.1097/APO.0000000000000271
- Singh P, Singh A. Choroidal melanoma. *Oman J Ophthalmol*. 2012;5(1):3-9. doi:10.4103/0974-620X.94718
- Shields CL, Dalvin LA, Ancona-Lezama D, Yu MD, Di Nicola M, Williams BK Jr, et al. Choroidal Nevus Imaging Features In 3,806 Cases And Risk Factors For Transformation Into Melanoma In 2,355 Cases: The 2020 Taylor R. Smith and Victor T. Curtin Lecture. *Retina*. 2019 Oct;39(10):1840-1851. doi: 10.1097/IAE.0000000000002440. PMID: 30608349.
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, IV: Local treatment Failure and Enucleation in the first 5 years after Brachytherapy. *Ophthalmology* 2002; 109: 2197-2206
- Chang MY, McCannel TA. Local treatment failure after globe-conserving therapy for choroidal melanoma. *Br J Ophthalmol* 2013; 97: 804-811.
- Correa R, Pera J, Gómez J, et al. 125I episcleral plaque brachytherapy in the treatment of choroidal melanoma: A single-institution experience in Spain. *Brachytherapy* 2009. 8; 290-296
- Kowal J, Markiewicz A, Dębicka-Kumela M, et al. Analysis of local recurrence causes in uveal melanoma patients treated with 125I brachytherapy - a single institution study. *J Contemp Brachytherapy*. 2019;11(6):554-562. doi:10.5114/jcb.2019.90985
- Dalvin LA, Shields CL, Ancona-Lezama DA, Yu MD, Di Nicola M, Williams BK Jr, et al. Combination of multimodal imaging features predictive of choroidal nevus transformation into melanoma. *Br J Ophthalmol*. 2019 Oct;103(10):1441-1447. doi: 10.1136/bjophthalmol-2018-312967. Epub 2018 Dec 6. PMID: 30523045.
- Shields CL, Shields JA, Gündüz K, Cater J, Mercado GV, Gross N, Lally B. Conjunctival melanoma: risk factors for recurrence, exenteration, metastasis, and death in 150 consecutive patients. *Arch Ophthalmol* 2000; 118: 1497
- Kaliki S, Shields CL. Uveal melanoma: relatively rare but deadly cancer. *Eye (Lond)* 2017; 31: 241-257.

24. Peddada KV, Sangani R, Menon H, Verma V. Complications and adverse events of plaque brachytherapy for ocular melanoma. *J Contemp Brachytherapy*. 2019;11(4):392-397. doi:10.5114/jcb.2019.87407
25. Collaborative Ocular Melanoma Study Group Incidence of cataract and outcomes after cataract surgery in the first 5 years after iodine 125 brachytherapy in the Collaborative Ocular Melanoma Study: COMS Report No. 27. *Ophthalmology*. 2007;114:1363–1371
26. Gunduz K, Shields CL, Shields JA, *et al.* Radiation Complications and Tumor Control After Plaque Radiotherapy of Choroidal Melanoma with Macular Involvement. *Am J Ophthalmol* 1999; 127: 579-589
27. Finger PT, Chin K. Anti-Vascular Endothelial Growth Factor Bevacizumab (Avastin) for Radiation Retinopathy. *JAMA Ophthalmol* 2007; 125(6):751–756 22
28. Kaliki S, Shields CL. Uveal melanoma: Relatively rare but deadly cancer. *Eye (Lond)* 2017;31:241-57.